Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2024 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

27 adaptive question sections   96 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section keeps the exact source demand beside practical working space. A source grid, graph, table, or diagram is retained once at a usable size.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 1; the selection contains 96 marks.

Contents

Question 01

4MA1-1H Q2

MODULAR UNIT 1

4 marks

2 450 students were asked how they travelled to school on Monday.
Each student walked or travelled by bus or travelled by car or travelled by bicycle.
Each student used just one method of travel.

One of these students is chosen at random.
The table shows information about the probability of each method of travel.

Method of travel	walk	bus	car	bicycle
Probability	0.20	x	$2x$	0.26

Work out how many of the 450 students travelled by car.

....

WORKING SPACE

Question 02
4MA1-1H Q6

MODULAR UNIT 1
2 marks

6 (a) Expand and simplify $(m + 5)(m - 8)$

....
(2)

WORKING SPACE

Question 03

4MA1-1H Q6

MODULAR UNIT 1

3 marks

(b) Solve $3n - 4 = \dfrac{5n + 6}{3}$

Show clear algebraic working.

$n =$ ....

(3)

WORKING SPACE

Question 04

4MA1-1H Q7

MODULAR UNIT 1
5 marks

7 $\mathscr{E} = \{23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}$
$A = \{\text{even numbers}\}$
$B = \{23, 29, 31\}$
$C = \{\text{multiples of } 3\}$

(a) List the members of the set

(i) $B \cup C$

....
(1)

(ii) $A' \cap C$

....
(1)

(b) Is it true that $B \cap C = \varnothing$?

Tick (✓) one of the boxes below.

Yes ☐ No ☐

Give a reason for your answer.

....

....
(1)

The set D has 4 members and is such that $D \cap (A \cup C) = \varnothing$

(c) List the members of set D

....
(2)

Working space for Question 04

Continue your method

UNIT 1
5 marks

WORKING SPACE

Question 05

4MA1-1H Q8

MODULAR UNIT 1
3 marks

8 A cylinder is placed on a table.

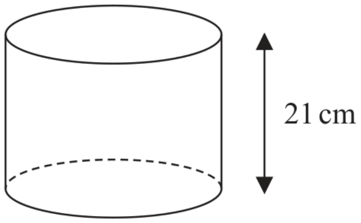


Diagram **NOT** accurately drawn

The volume of the cylinder is 1575 cm^3
The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

.... newtons/cm^2

WORKING SPACE

Question 06

4MA1-1H Q10

MODULAR UNIT 1

4 marks

10 (a) Simplify $(2p)^0$ where $p > 0$

....

(1)

$y^9 \times y^{-3} = y^n$

(b) Find the value of n

$n =$ ....

(1)

(c) Simplify fully $(5a^4c^2)^3$

....

(2)

WORKING SPACE

Question 07

4MA1-1H Q11

MODULAR UNIT 1

4 marks

11 The diagram shows a roof support.

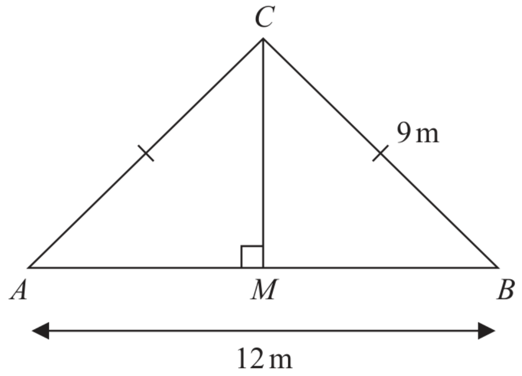


Diagram **NOT** accurately drawn

The roof support is made from four lengths of wood, AB, AC, BC and MC

$AC = BC = 9$ m $\quad AB = 12$ m

angle $AMC = 90°$

Lewis is going to buy lengths of wood to make the roof support.

The wood costs 21.50 euros per metre.
Each length of wood he buys has to be a whole number of metres.

Work out the total cost of the wood Lewis needs to buy.
Show your working clearly.

.... euros

Working space for Question 07

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 08
4MA1-1H Q12

MODULAR UNIT 1
2 marks

12 (a) Factorise fully $6y^2 - 5y - 4$

....
(2)

WORKING SPACE

Question 09

4MA1-1H Q12

MODULAR UNIT 1

3 marks

(b) Express $\dfrac{2x+1}{4x} + \dfrac{7-5x}{3x}$ as a single fraction in its simplest form.

....

(3)

WORKING SPACE

Question 10
4MA1-1H Q13

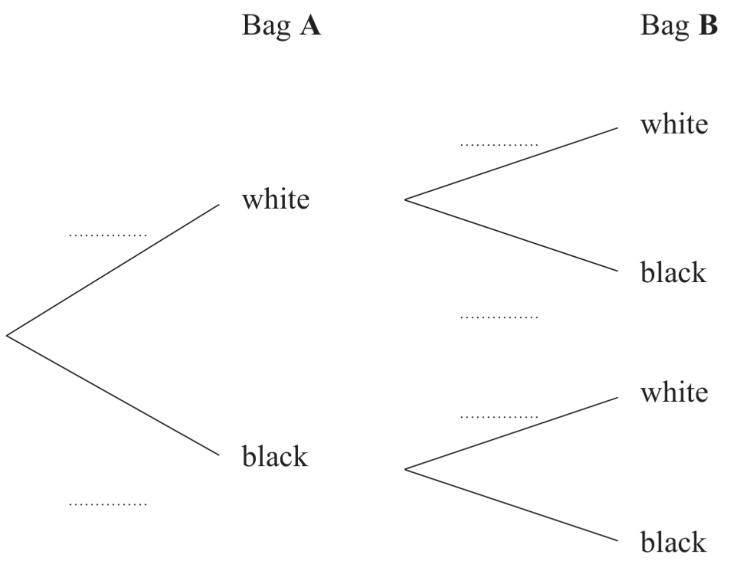
MODULAR UNIT 1
5 marks

13 Harman has two bags of beads.

In bag **A**, there are 3 white beads and 7 black beads.
In bag **B**, there are 5 white beads and 4 black beads.

Harman takes at random a bead from bag **A** and a bead from bag **B**

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that Harman takes two beads of the same colour.

....

(3)

Question 11

4MA1-1H Q16

MODULAR UNIT 1
4 marks

16 There are 20 sweets in a box.

15 of the sweets are red
5 of the sweets are yellow

Fred takes at random 3 sweets from the box.

Work out the probability that Fred takes at least one sweet of each colour from the box.

....

WORKING SPACE

Question 12
4MA1-1H Q17

MODULAR UNIT 1
3 marks

17 Show that $\dfrac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a+\sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

WORKING SPACE

Question 13

4MA1-1H Q19

MODULAR UNIT 1
5 marks

19 Here is quadrilateral *ABCD*

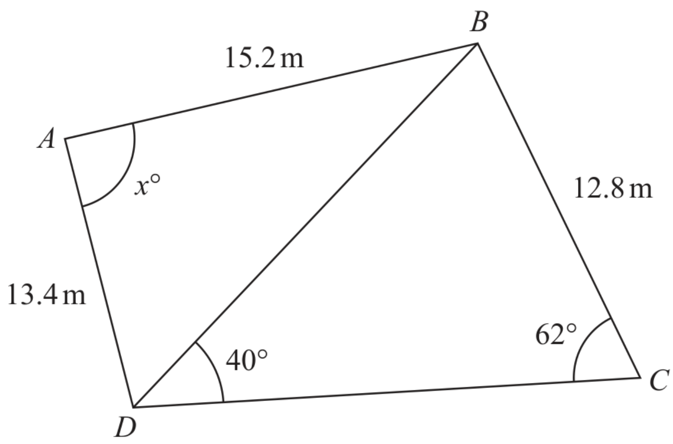


Diagram **NOT** accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

$x =$ ....

WORKING SPACE

Question 14

4MA1-1H Q20

MODULAR UNIT 1

4 marks

20 The diagram shows a sector $OABC$ of a circle centre O

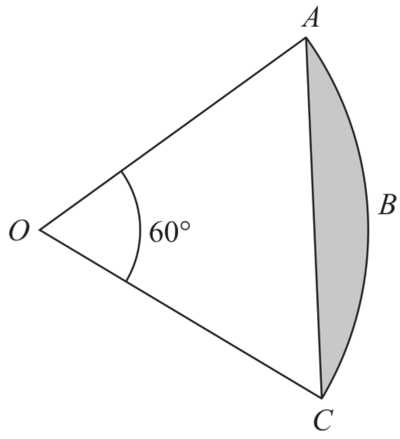


Diagram **NOT** accurately drawn

Angle $AOC = 60°$

The area of the shaded segment ABC is $38\,\text{cm}^2$

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

.... cm

Working space for Question 14

Continue your method

UNIT 1
4 marks

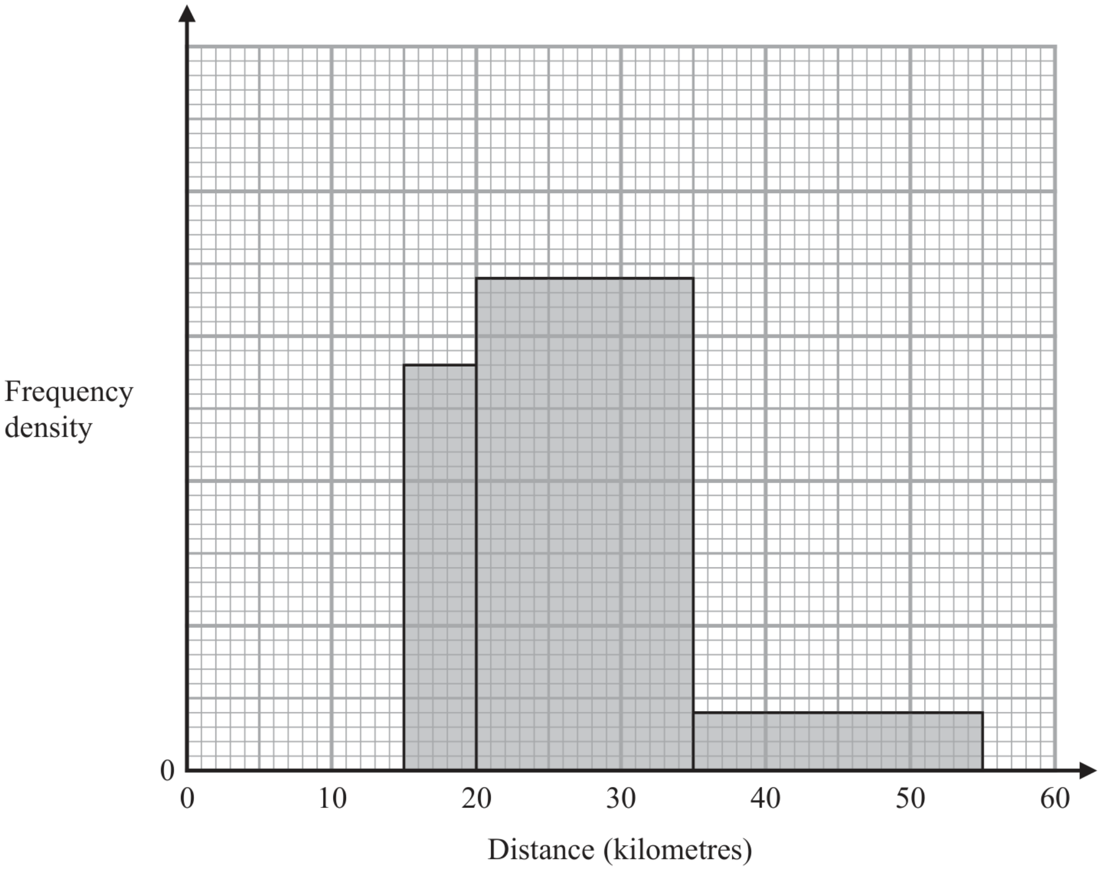
WORKING SPACE

Question 15

4MA1-1H Q22

MODULAR UNIT 1
3 marks

22 The incomplete histogram shows some information about the distances, in kilometres, that 100 adults ran last week.



All of the adults ran at least 5 kilometres.
None of the adults ran more than 55 kilometres.
14 adults ran between 15 kilometres and 20 kilometres.

Complete the histogram.

Question 16

4MA1-1H Q25

MODULAR UNIT 1
4 marks

25 $f(x) = 17 - 3x^2 + 12x$

Write $f(x)$ in the form $a - b(x - c)^2$ where a, b and c are constants.

$f(x) =$ ....

WORKING SPACE

Question 17
4MA1-2H Q3

MODULAR UNIT 1
3 marks

3 A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

.... km

WORKING SPACE

Question 18

4MA1-2H Q4

MODULAR UNIT 1
3 marks

4 Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

WORKING SPACE

Question 19

4MA1-2H Q5

MODULAR UNIT 1

3 marks

5 The diagram shows triangle *ABC*

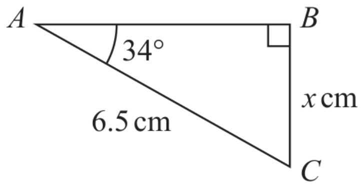


Diagram **NOT** accurately drawn

Work out the value of x

Give your answer correct to one decimal place.

$x =$ ....

WORKING SPACE

Question 20

4MA1-2H Q6

MODULAR UNIT 1
3 marks

6 Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

.... kilometres per hour

WORKING SPACE

Question 21

4MA1-2H Q7

MODULAR UNIT 1
4 marks

7 The diagram shows a 6-sided shape *ABCDEF*

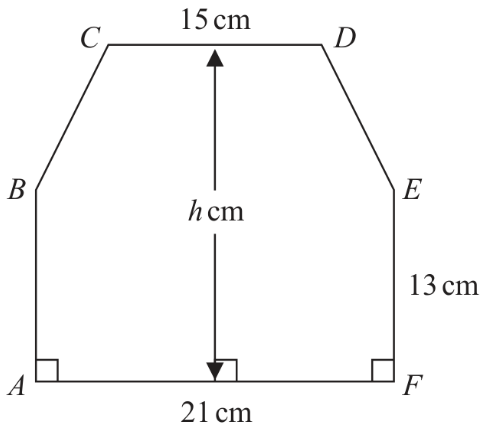


Diagram **NOT** accurately drawn

$AF = 21$ cm $\quad CD = 15$ cm $\quad AB = FE = 13$ cm

The perpendicular height of the shape is h cm
CD is parallel to AF

The area of the shape is 390 cm^2

Work out the value of h

$h =$ ....

Working space for Question 21

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 22

4MA1-2H Q11

MODULAR UNIT 1

3 marks

11 (i) Factorise $x^2 + 9x - 22$

....

(2)

(ii) Hence, solve $x^2 + 9x - 22 = 0$

....

(1)

WORKING SPACE

Question 23

4MA1-2H Q14

MODULAR UNIT 1
4 marks

(b) Solve $\dfrac{2x+3}{5} + \dfrac{6x-5}{4} = \dfrac{163}{100}$

Show clear algebraic working.

$x =$ ....

(4)

WORKING SPACE

Question 24

4MA1-2H Q16

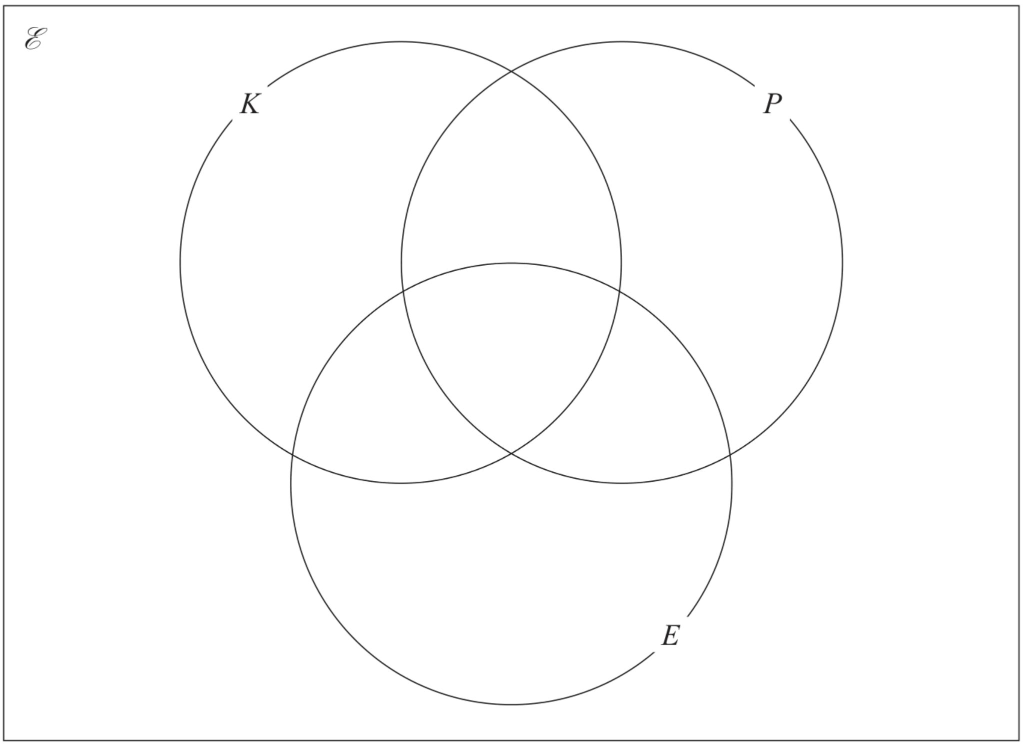
MODULAR UNIT 1
7 marks

16 60 art students were asked if they would like to attend workshops for knitting (K), for photography (P) or for embroidery (E)

Of these students

9 chose knitting, photography and embroidery
17 chose knitting and photography
16 chose photography and embroidery
20 chose knitting and embroidery
28 chose photography
39 chose embroidery
2 chose none of the workshops

(a) Using this information, complete the Venn diagram to show the numbers of students in each subset.



(3)

Question 24 (continued)

4MA1-2H Q16

MODULAR UNIT 1

7 marks

One of the students is chosen at random.

Given that this student chose photography,

(b) find the probability that this student also chose knitting.

....
(2)

(c) Find $n(P \cap K')$

....
(1)

(d) Find $n([P \cup E] \cap K)$

....
(1)

Question 25

4MA1-2H Q18

MODULAR UNIT 1

2 marks

18 The straight line **P** has equation $5y + 2x = 7$

Find the gradient of a straight line that is perpendicular to **P**

....

WORKING SPACE

Question 26

4MA1-2H Q19

MODULAR UNIT 1

3 marks

19 $G = \dfrac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number
$f = 6.62$ correct to 2 decimal places
$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G
Give your answer correct to 3 decimal places.
Show your working clearly.

....

WORKING SPACE

Question 27

4MA1-2H Q23

MODULAR UNIT 1

3 marks

23 Simplify $$\frac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

....

WORKING SPACE